

Two math games, six questions

MTH 408 Independent Study, Fall 2026, Week 5

Numberle (numberle.org) asks you to guess a hidden equation in six tries, Wordle-style. Reach hands you a few digits and a target and asks how many ways there are to get there, with a class working one list together. The questions in the left column are the assessment tool. A check means yes, $\frac{1}{2}$ means partly, blank means no. The last column is for your own game.

Question	What we're looking for	Numberle	Reach	Yours
What do you get better at?	Expert play needs mathematical thinking, not just symbol tracking		✓	
	Arithmetic is load-bearing (you can't play without computing)	$\frac{1}{2}$	✓	
What does the feedback know?	Feedback responds to the value or structure of a guess, not its characters		✓	
Where's the argument?	A puzzle has more than one right answer		✓	
	Ends in a claim a student can defend ("we have them all"); the count stays hidden until then		✓	
What counts as the same?	The game's rule for "same answer" matches the mathematics		✓	
	That rule is written where students can read and argue with it		✓	
Who authors the puzzles?	Teacher can set the digits, target, and rules		✓	
	Students can pose puzzles or change a rule and see what breaks			
What breaks across grade bands?	Something mathematical changes when you move it up or down a band		✓	
	Same interface at every band; only the rules differ		✓	
Classroom use	A class can work one puzzle together, in class or overnight		✓	
	Designed to end: a closing time, then a discussion		✓	
	The fastest student can't fill the board alone		✓	
	No streaks, share strings, daily-return hooks, ads, or cross-promotion		✓	
Where both do fine	Learn it in under a minute; works on a phone	✓	✓	
	Same daily puzzle for everyone	✓	✓	
	Runs with no setup	✓	$\frac{1}{2}$	

The one-line critique of Numberle. Its own rules say " $a+b=c$ and $b+a=c$ are different answers." The game has to say that, because it is a string-matching game, and string matching can't afford commutativity. Everything else in the table follows from that.

Where Reach is still weak. The authoring row: a teacher can set a puzzle, a student can't. And "gets better at math" is an argument, not a measurement; nobody has checked whether estimating in this game transfers anywhere. Treat both checks in that first row as claims to test.

How to use this. Put your game in the last column, honestly. Then pick one blank and ask: what is the smallest change that would earn the check?